

One-Time Pads in Cryptography

Learn how to use one-time pads, how to set up secure one-time pad communications and how to deal with its various security issues



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Information security means protecting information and information systems from unauthorised access, use, disclosure, disruption, modification, perusal, inspection, recording or destruction. It is mainly concerned with the confidentiality, integrity and availability of data regardless of the form (electronic, print or other) the data may take.

One-time pad encryption is a basic yet solid method in information security which can be used for securing short and long messages. One-time pad is a cryptosystem where the plaintext is combined with a random key. This key or set of keys is called a pad.

If properly used, the system provides truly unbreakable encryption and it will be impossible for any eaves-

dropper to decrypt or break one-time pad encrypted message by any type of cryptanalytic attack without the proper key, even with infinite computational power.

One-time pad cryptosystem has a property of perfect secrecy. In perfect secrecy, the ciphertext gives absolutely no additional information about the plaintext. Thus, the *a priori* probability of a plaintext message M is the same as the *a posteriori* probability of a plaintext message M , given the corresponding ciphertext. In fact, all plaintexts are equally probable. This is a strong notion of cryptanalytic difficulty.

It is easy to learn to work with one-time pads. The system is transparent, and one does not need any special equipment or any knowledge about cryptographic technique. Also, it is possible to integrate various other

security techniques to strengthen the system.

History

One-time pad is also called Vernam cipher or the perfect cipher. Used by special operations teams and resistance groups in Second World War, it became popular with intelligence agencies and their spies during the Cold War and beyond. One-time pad gained reputation as a simple but solid encryption system in the intriguing world of intelligence, espionage, covert operations, and military and diplomatic communications, with an absolute security which is unmatched by modern crypto algorithms.

The story of one-time pad starts in 1917, when AT&T research engineer Gilbert Vernam developed a system to encrypt teletype TTY communications. He mixed a five-bit Baudot-coded punched paper tape, containing the message, with a second punched paper tape, the key, containing random five-bit values. To mix the punched tapes, a modulo 2 addition (later commonly known as the boolean XOR or Exclusive OR) was performed with relays, and the key tape ran synchronously on the sending and receiving telex machine. It was the first automated direct online encryption system.

Vernam realised that encryption with short key tapes (basically a poly-alphabetic cipher) would not provide enough security. Initially, he used a mix of two key tape loops, with relatively prime length, creating one very long random key.

Later Captain Joseph Mauborgne showed that even the double key tape system could not resist cryptanalysis if large volumes of message traffic were encrypted. Mauborgne concluded that